Julien Petit--Colboc

Computer Science Engineering Student – AI, VR & Robotics
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Rouen, France
Portfolio – sliiz.com

Summary

4th-year Computer Science engineering student (work-study) at CESI, apprentice at CAPFI since January 2026. Main developer of Cerebro, an LLM-based matching system now in production, where I worked on making scores explainable and measurable. Previously a research intern at CESI LINEACT on VR teleoperation of robots. Seeking an internship abroad from June 2027 (12 to 17 weeks) in AI, VR or robotics, ideally within a research team.

Profiles

GitHub
SliiZ-code

LinkedIn
julien-petit-colboc

Languages

French
Native
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English
Professional, TOEIC scheduled
●●●●○

German
Beginner
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Skills

AI & LLM
LLM pipelines, Prompt engineering, Structured outputs, OpenAI & Gemini APIs, Stable Diffusion, Whisper

Languages
Python, C#, C++, Dart, JavaScript

Backend
FastAPI, .NET, REST APIs, SQLite, Pydantic

VR, 3D & Hardware
Unity, Arduino

DevOps & Systems
Docker, GitLab CI, Git Flow, Linux/WSL, VMware, Wireshark

Interests

Running (half-marathon, March 2027), calisthenics, graphic design, travel

Experience

CAPFI — Paris
Software Engineer Apprentice (R&D, AI) — Jan 2026 – Present
Main developer of Cerebro, an internal tool that matches consultant profiles to client requests for proposals (RFPs) using LLMs.

- Built an AI pipeline that extracts, categorises and scores RFP × profile pairs, with reliable asynchronous batch processing; deployed in production after processing several hundred RFPs.
- Calibrated the scoring prompt (temperature, required vs. desired skills, three sub-scores) and added short match justifications so users can check each score.
- Started an annotated reference set to measure false positives and false negatives.
- Built the daily report sent to about 10 sales staff; secured the API and web interface with authentication and SSO. *Stack: Python, FastAPI, Pydantic, SQLite, Docker, GitLab CI, Gemini.*

CESI LINEACT — Rouen
Research Intern — Apr 2025 – Jul 2025
Built a VR app for remote robot teleoperation (Scrum; C#, .NET, Unity, REST APIs).

- Streamed a 360° camera mounted on the robot into an immersive VR view.
- Secured robot access with AES-256 encryption.
- Designed a real-time path interface to add, edit or delete waypoints.

Projects

Blink — 2024
Python pipeline that generates short videos from a single topic: GPT-3.5 writes the script, Stable Diffusion XL creates the images, OpenAI TTS records the voice-over and Whisper generates the subtitles. Asynchronous API calls, automated editing with MoviePy.

Thrive — 2024
Cross-platform task manager (Android, iOS, Windows, macOS, Web) built with Flutter/Dart: local Isar database, categories, notifications.

Education

CESI École d'Ingénieurs — Engineering degree (Master's level)
Computer Science, work-study — Rouen • Oct 2025 – Sep 2028

CESI — Integrated preparatory cycle
Computer Science — Rouen • Sep 2023 – Jul 2025

- Conway's Game of Life in C++ (pair project, main contributor): MVC architecture, console and SFML graphical modes, toroidal grid, obstacle cells.

Activities

BDE CESI Rouen – Head of Communication — Oct 2025 – Present
Lead the student union's communication: Instagram content, event posters, photo and video coverage.

CESI Robotics Club — 2023 – 2024
Built the remote control of a BB-8 replica: web joystick (HTML, JS) and a Flask server driving the Arduino over serial.